Operating Systems
Homework 3 (HW3)

Instructor: Dr. Probir Roy

Instructions

- No teamwork is allowed for all homework assignment.
- `VeriCite` option has been enabled on Canvas to compare your submission with other submissions from this semester and past semesters and Internet database.
- Your solution should be written in a ".docx", ".doc", or ".pdf" format and submitted to the assignment folder HW3 on Canvas.

Questions

1. Describe scenarios to show that multi-process is better than multi-thread and vice versa.
2. As we discussed in the class, if two threads execute the same instruction `counter++`, where `counter` is a shared variable, then a race condition happens. Explain what happens exactly.
3. Explain the difference between the methods of yielding and sleeping used in overcoming spinning waiting. Which one is more efficient? Why?
4. Explain how the atomic operation `TestAndSet` is used to implement lock.
5. Explain how lock is implemented in `xv6` and how it can be used to avoid data inconsistency.
6. A resource is to be shared among different threads, each of which is associated with a priority. Whenever a new thread tries to access the resource, it will be admitted if either (i) its priority is higher than the priority of one of the threads currently accessing the resource or (ii) there is no any other threads currently accessing the resource. Use mutex and condition variable to coordinate threads' access to the resource. Write it in a pseudo code format.
7. In Gallap Park at Ann Arbor, there is one lane bridge. Vehicles coming from the northeast and the southwest arrive at this one-lane bridge. Vehicles heading in the same direction can cross the bridge at the same time, but vehicles heading in opposite directions cannot. And, if there are ever more than `two` vehicles on the bridge at one time, the bridge will collapse under their weight. If a vehicle arrives at the bridge, the number of vehicles on the bridge is less than two and they all head in the same directions, this new vehicle should be allowed to cross the bridge without waiting. Use semaphore to coordinate vehicles' crossing the one-lane bridge. Write it in a pseudo code format .

8. We consider the example about two threads accessing a printer and a plotter in class (check the slides `2-4 concurrency-deadlock.pptx`). We consider the case that at Position T no deadlock avoidance is applied and then the system will enters the red-bordered state. Draw the Resource Allocation Graph (RAG) after the system enters the red-bordered state.

9. Explain how to prevent deadlock.

10. Explain the difference between deadlock prevention and deadlock avoidance.